

Machine Learning: Building a Predictive Model *with* Scikit-learn



Scikit-learn was initially developed by David Cournapeau as a Google Summer of Code project in 2007. The project now has more than 30 active contributors with paid support from Inria, Google, Tinyclues and the Python Software Foundation. Scikit-learn provides algorithms for supervised and unsupervised learning, using a Python interface.

Machine learning is one of the branches of computer science in which algorithms (running inside computers) learn from the data available to them. With this learning mechanism, various predictive models can be arrived at.

The machine learning systems (or algorithms) can be broadly classified into many categories, based on various factors and methods. The most popular of these are:

- Supervised learning
- Unsupervised learning
- Semi-supervised learning
- Reinforcement learning
- Online/batch learning

Supervised learning: In supervised learning, the data we input to these algorithms also includes the expected solution, which is called labels.

The supervised learning algorithm consists of sets of variables called dependent and independent variables. Dependent variables are derived from the target or outcome and independent variables are predictors.

Supervised learning occurs when the machine is taught or trained using data that is well labelled (which means that some data is already tagged with the correct answer or classification). Once this is done, the machine is provided with a new set of data or examples, so that this algorithm analyses the training data (set of training examples or data sets) and produces the desired outcome from labelled data.

The most popular and talked about supervised learning algorithm is the 'Spam or Ham' classification which is used for spam email classification. Another popular algorithm relates to fruit classification — if the shape of an object is round with a depression at its top and the colour is red, then it will probably be labelled as 'apple'. And if the shape of an object is a long, curved cylinder that is green-yellow in colour, then it will probably be labelled as 'banana'.

Some of the most important and heavily used supervised learning algorithms are:

- k-Nearest neighbours
- Linear regression
- Logistic regression